SAMUEL OJIMA-OJO ALIKALI

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Portfolio: samuelalikali02-glitch.github.io/#projects | GitHub: github.com/samuelalikali02-glitch

PROFESSIONAL PROFILE

Final-year Mechatronics Engineering student at the Federal University of Technology, Minna, with practical exposure to embedded systems, IoT, electronics, CAD, automation and technical project development. Currently undertaking industrial training at NASENI while building skills in Web3/blockchain and introductory AI/ML. Interested in combining engineering, software and connected technologies to develop practical solutions.

EDUCATION

B.Tech. Mechatronics Engineering — Federal University of Technology, Minna (FUTMINNA)

Five-year programme | Degree in progress | Final year

Federal Government College, Kwali — Secondary education, 2018–2020

INDUSTRIAL TRAINING

Industrial Trainee — National Agency for Science and Engineering Infrastructure (NASENI), Abuja

Directorate of Engineering and Manufacturing Infrastructure | May 2026–Present

- Gain practical exposure to engineering and manufacturing infrastructure activities.
- Support technical tasks, equipment-related activities and engineering documentation.
- Develop practical understanding of engineering systems, workplace procedures and project workflows.
- Recently assigned to contribute to an existing bean-picker project; activities are ongoing.

SELECTED PROJECTS

Ethereum Block Explorer | Web3 Project

Built and deployed a web application for searching Ethereum block numbers, transaction hashes and wallet addresses, with wallet balance display and Etherscan navigation. Uses React/JavaScript and Alchemy-based blockchain data access; deployed on Vercel.
Live demo: ethereum-block-explorer-two.vercel.app
Source: github.com/samuelalikali02-glitch/ethereum-block-explorer

Arduino-Based 3-DOF Flight Simulator / Servo Mechanism

Worked with an Arduino Nano and servo hardware to explore motion control and mechanical actuation; tested sketches and investigated servo twitching, movement and power-supply issues.

Embedded Systems and IoT Prototypes

Explored Arduino/ESP32 microcontrollers, sensors, actuators and connectivity for automation and IoT concepts, including smart energy monitoring and power-sharing.

Autonomous Emergency Surveillance UGV

Contributed to concept development involving mobile robotics, sensing and intelligent surveillance for high-risk environments.

Smart Water Dispenser

Worked on an automated dispensing concept involving sensor integration, control and component planning.

CAD Heat Exchanger Design

Completed engineering design coursework involving shell-and-tube and plate heat exchanger components, modelling and assembly considerations using SolidWorks.

UAV Propulsion / Motor Project Exposure

Additional project exposure related to UAV motor/propulsion concepts; technical work and project stage should be described in detail when confirmed.

Web Development Portfolio

Created a personal portfolio using foundational HTML and CSS to present projects and technical interests. View portfolio.

TECHNICAL SKILLS

Embedded Systems & IoT	Arduino, ESP32, sensors, actuators, Wi-Fi/Bluetooth, IoT fundamentals
Programming & Web	C++, Java, JavaScript, HTML/CSS, Arduino IDE
Web3 / Blockchain	Ethereum fundamentals, Alchemy SDK, blockchain data access, smart-contract learning
CAD & Engineering	SolidWorks, mechanical design, engineering drawings, electronics troubleshooting, control and automation fundamentals
AI/ML	Foundational learning and introductory experimentation; developing skills for potential engineering and IoT applications

WEB3 & AI/ML DEVELOPMENT

Expanding from embedded engineering into Web3 through JavaScript, Ethereum and smart-contract learning. Also building foundational AI/ML knowledge, with a long-term interest in applying intelligent systems to IoT and engineering projects, including the MakerChain concept.

CORE STRENGTHS

Problem solving • Teamwork • Technical communication • Adaptability • Research • Project development • Quick learning • Engineering innovation

References available on request.